

CYBERSECURITY COURSE — WALAILAK UNIVERSITY

Introduction to Cybersecurity

Principles of Digital Defense

Lecture by Assist. Prof. Dr. Chanankorn Jandaeng

The Stakes of the Digital Battlefield

\$10 Trillion The expected global cost of cybercrime by 2025.

Core Definition

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks, damage, or unauthorized access.

The Assets We Protect



Critical Infrastructure



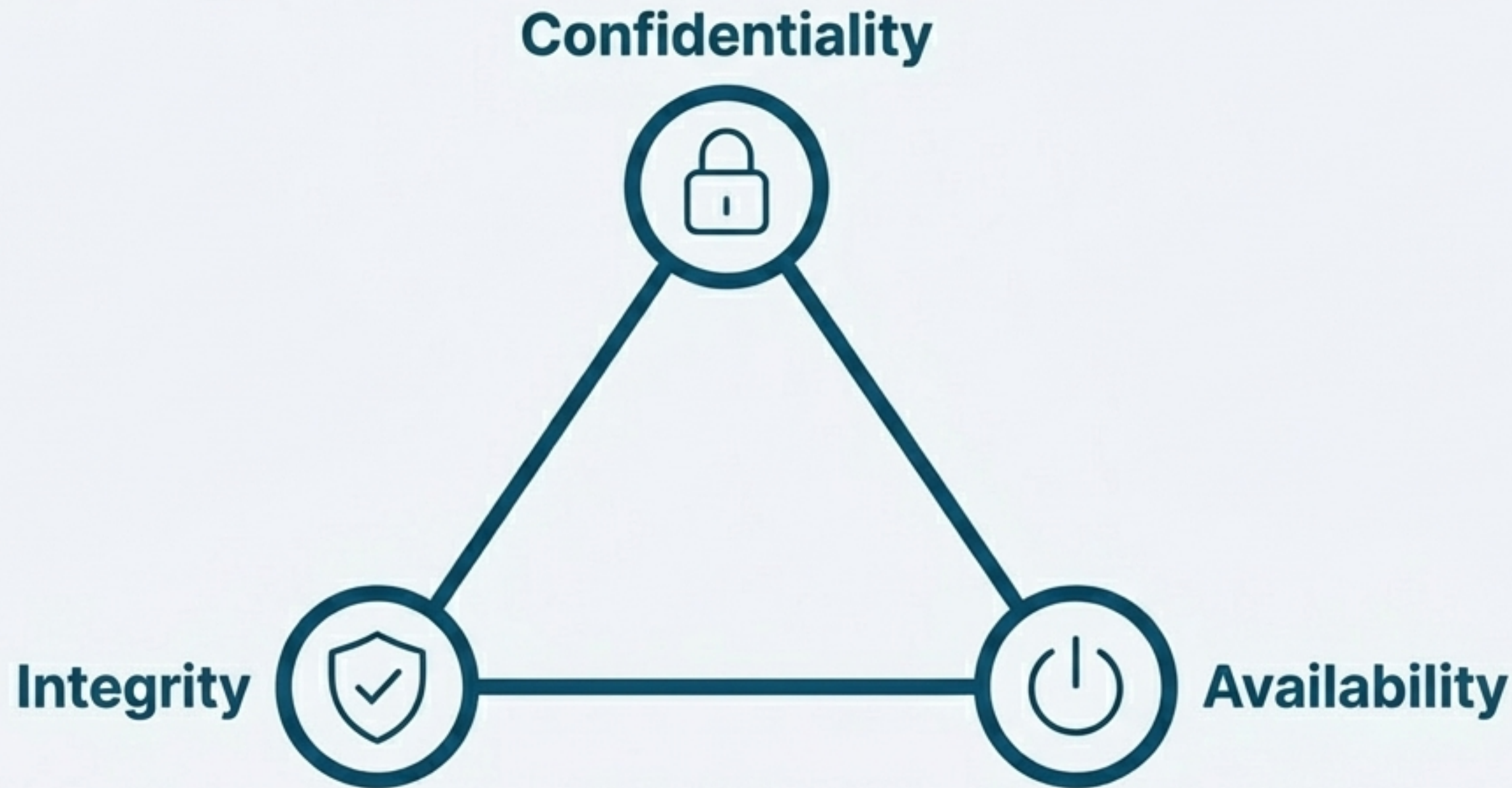
Personal Data



National Security

The Defender's Mandate: The CIA Triad

Our defensive strategy is built on three core principles. Together, these define the goals of all cybersecurity mechanisms.



The Three Pillars of Digital Defense



Confidentiality

Definition

Protecting information from unauthorized disclosure.

Mechanisms

Encryption, Access control, Data classification.

Example

Ensuring only authorized users can view patient medical records.



Integrity

Definition

Safeguarding data from unauthorized modification or destruction.

Mechanisms

Checksums, hashing, Digital signatures, Audit logs.

Example

Detecting unauthorized changes in financial transaction data.



Availability

Definition

Ensuring information and systems are accessible when needed.

Mechanisms

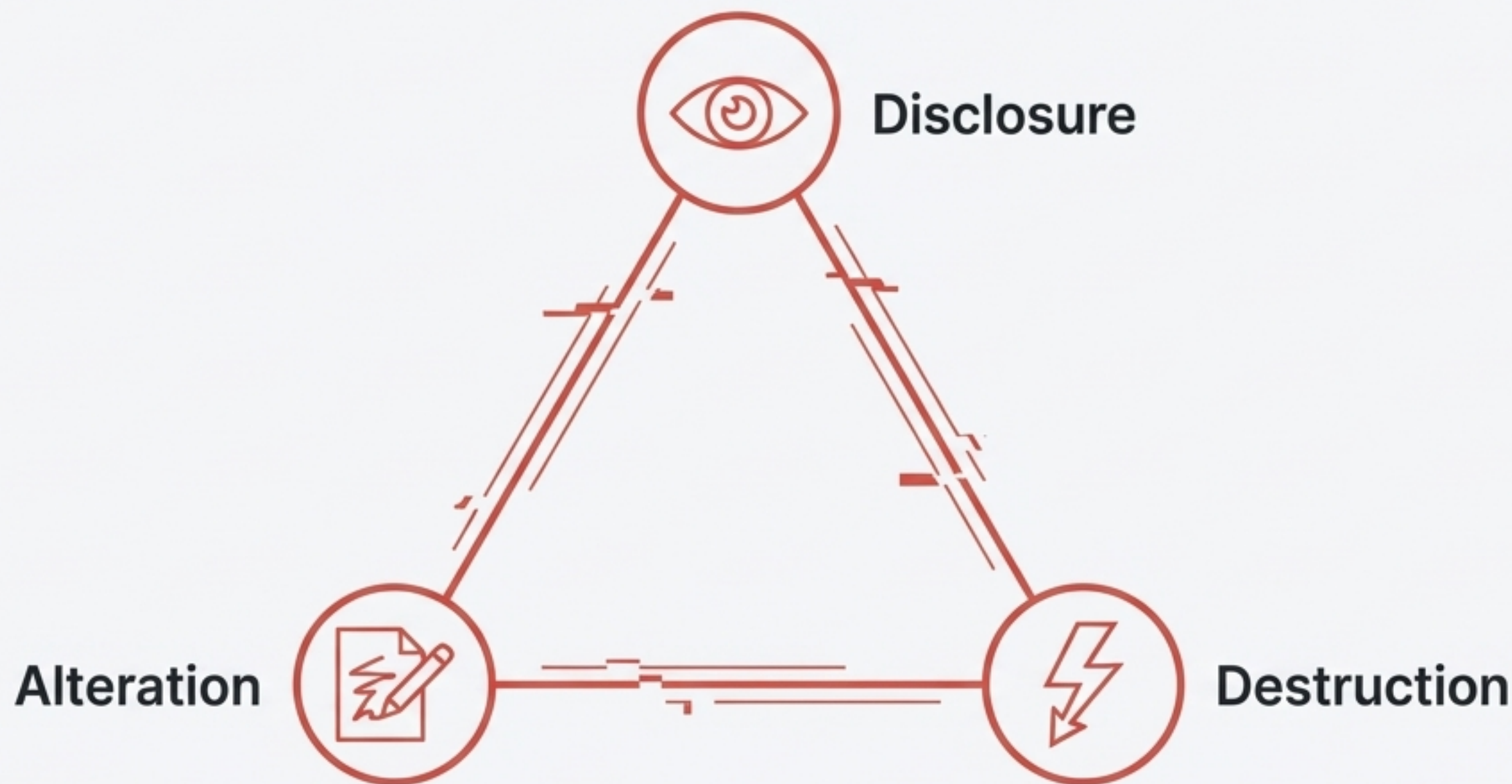
Redundancy and backups, DDoS mitigation, Fault tolerance.

Example

A banking system must stay online for 24/7 customer transactions.

The Adversary's Playbook: The DAD Triad

To build a strong defense, we must understand the attacker's objectives. **The DAD Triad** represents **the opposite** of the **CIA Triad** and helps us anticipate their strategies.



Anatomy of an Attack: Disclosure, Alteration, Destruction



Disclosure

Definition

Unauthorized exposure of confidential information.

Impact

Violates **Confidentiality**.

Example Attacks

Data breaches, Phishing, Eavesdropping.



Alteration

Definition

Unauthorized modification of data or system configurations.

Impact

Targets **Integrity**.

Example Attacks

SQL injection, Malware changing log files, DNS poisoning.



Destruction

Definition

Irreversible deletion or corruption of data or systems.

Impact

Compromises **Availability**.

Example Attacks

Ransomware, Logic bombs, Physical destruction.

The Core Conflict: A Perpetual Struggle

Effective security is about anticipating and detecting DAD behaviors in order to maintain CIA objectives.



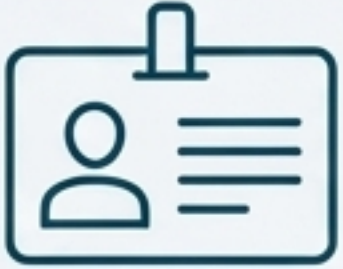
Key Insight: Network monitoring systems are designed to **detect DAD behaviors** to maintain **CIA objectives**.

The Rules of Engagement: The AAA Framework

The AAA model—Authentication, Authorization, and Accounting—provides the operational functions for enforcing security. It's *how* we implement our CIA strategy.



AAA in Action: From Identity to Audit Trail



Authentication

Purpose

Verify identity.

Methods

- Passwords, Multi-Factor Authentication (MFA), Biometrics, Digital certificates.



Authorization

Purpose

Determine what an authenticated entity can access.

Mechanisms

- Role-Based Access Control (RBAC), Access Control Lists (ACLs).



Accounting

Purpose

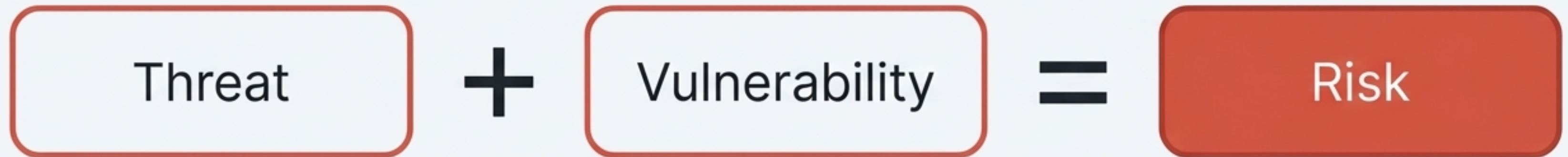
Track and record actions for auditing, compliance, and forensics.

Data Collected

- Login attempts, commands executed, resources accessed.

Understanding the Battlefield: Threat, Vulnerability, and Risk

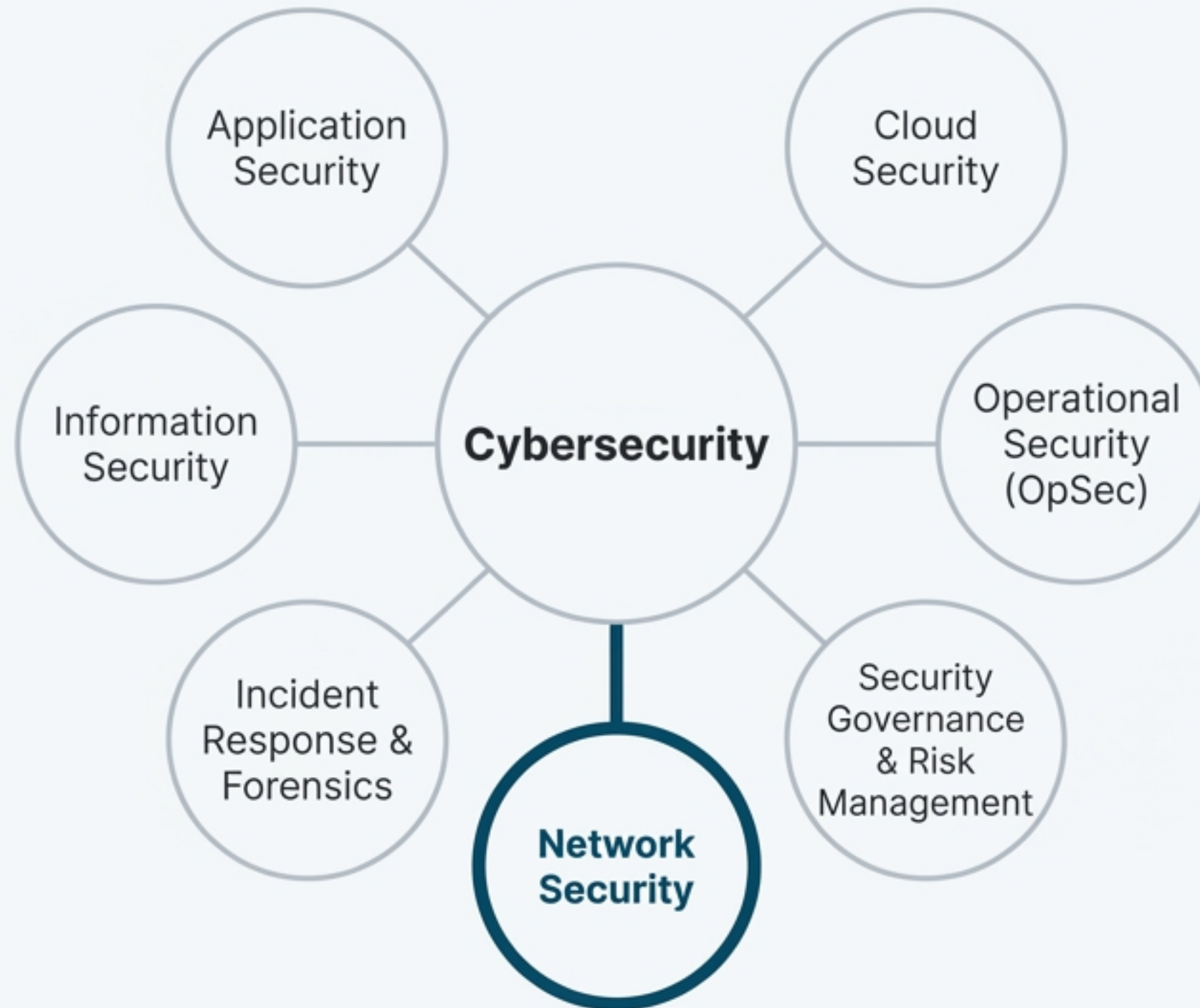
An attacker's threat exploits a vulnerability, creating a risk to our assets.



Threat	The potential cause of harm. (e.g., A hacker, malware).
Vulnerability	The weakness or flaw that is exploited. (e.g., Unpatched software, weak passwords).
Risk	The probability and impact of the harm.

Theaters of Operation: Major Cybersecurity Domains

Cybersecurity is a vast field with specialized domains, each protecting a different aspect of the digital ecosystem.



Our Mission: Securing the Network

Network monitoring is our primary intelligence-gathering tool, giving us the visibility to detect threats in real-time by turning raw data into actionable intelligence.

Network Security Focus

Protecting network infrastructure and data in transit.

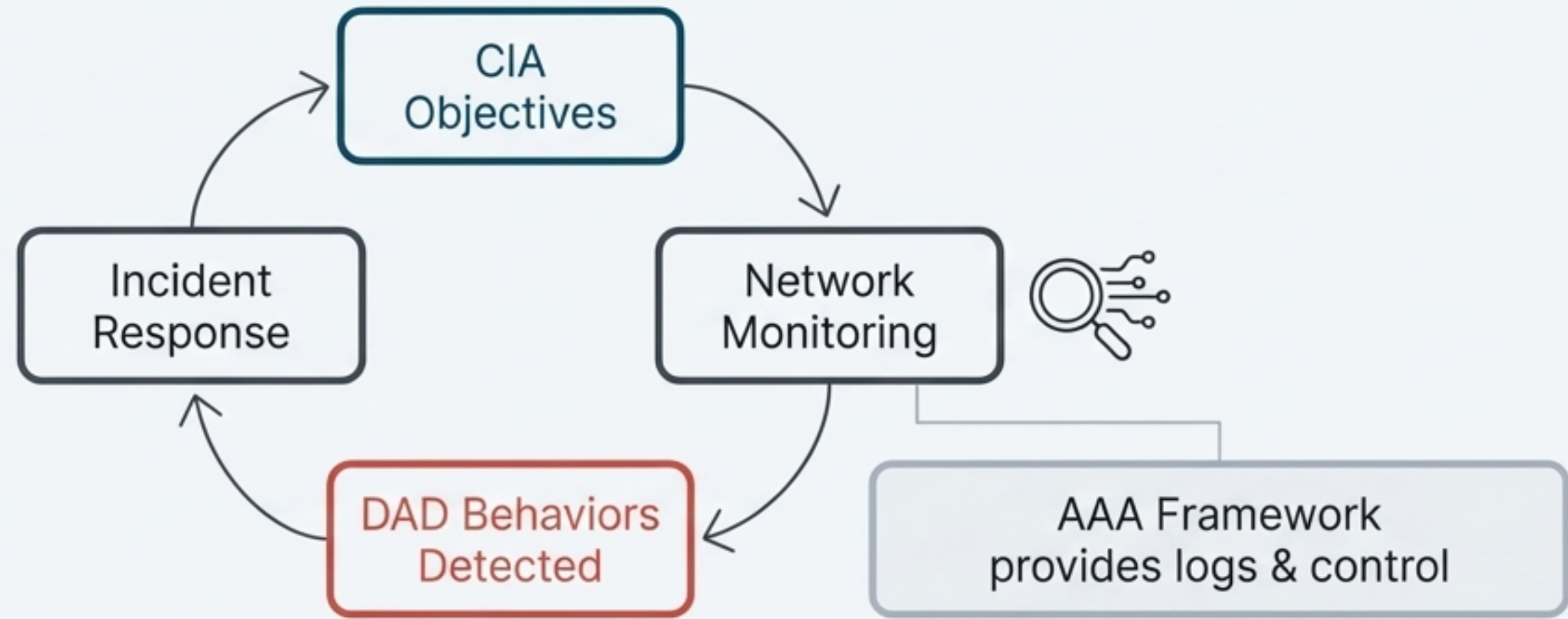
Key Monitoring Tools & Techniques

- SIEM (Security Information and Event Management)
- IDS/IPS (Intusion Detection/Prevention Systems)
- NetFlow and Packet Capture



The Unified Defense Model in Practice

Our understanding of **CIA** vs. **DAD**, enforced by **AAA**, is what allows us to design effective network monitoring systems.



To Uphold **Confidentiality**... we monitor for **Disclosure** behavior (e.g., detecting unusual outbound traffic).

To Uphold **Integrity**... we monitor for **Alteration** behavior (e.g., identifying log anomalies or unauthorized changes).

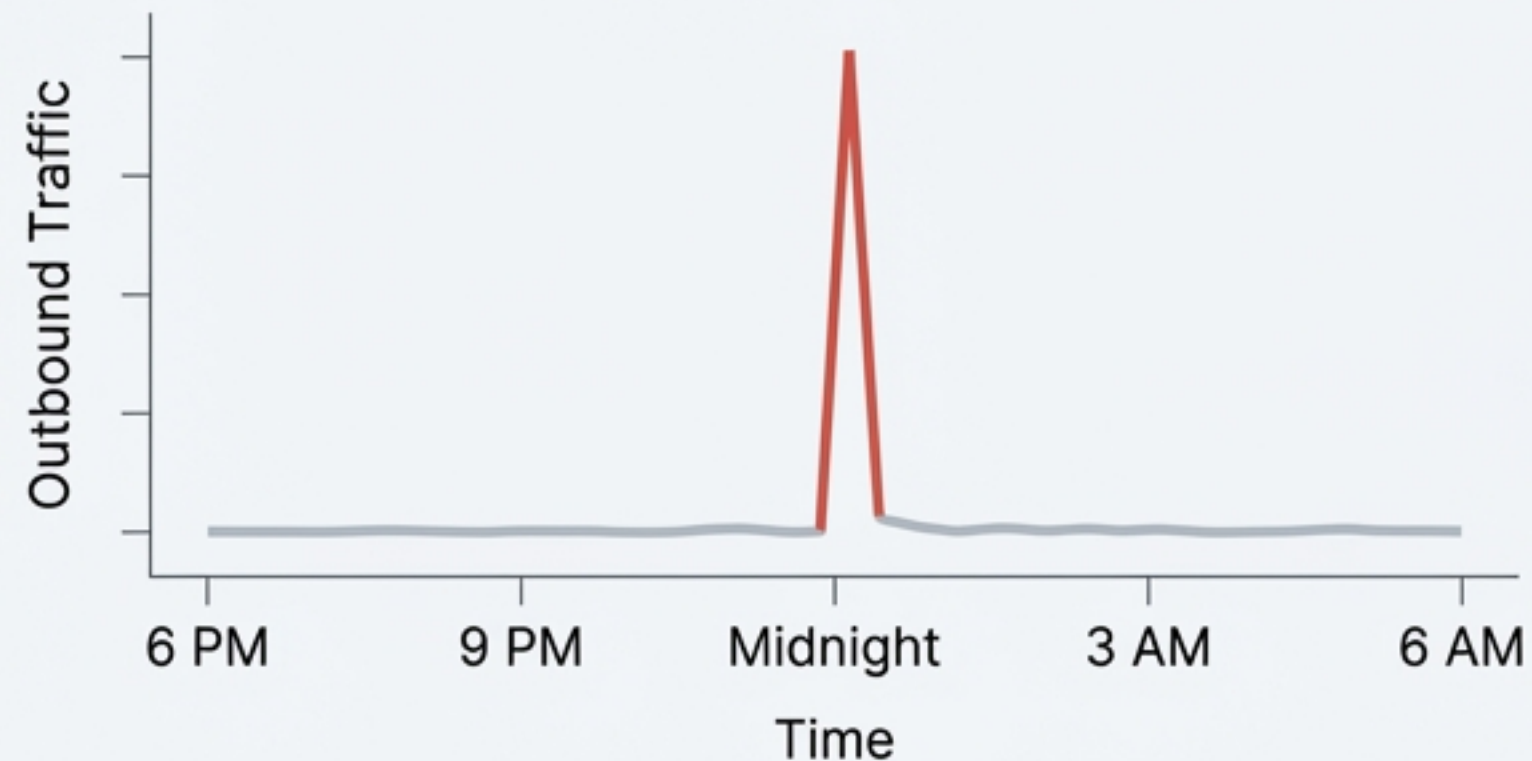
To Uphold **Availability**... we monitor for **Destruction** behavior (e.g., detecting DDoS traffic patterns).

Your Turn: Analyze the Anomaly

Apply these principles to a real-world scenario.

Scenario

A university network server shows massive, unusual outbound traffic to an unknown IP address at midnight.



Questions for the Analyst

1. Which CIA principle is at risk? What DAD activity might this be?
2. What AAA logs would be your first step to investigate?
3. What monitoring tool (SIEM, IDS) might have triggered the initial alert?

Mission Briefing Complete: Core Mental Models

- ✓ The **CIA Triad** defines our defensive objectives.
 - ✓ The **DAD Triad** reveals attacker motivations.
 - ✓ The **AAA Framework** provides operational control.
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Forward Look

These principles are the foundation for designing intelligent network monitoring systems that can effectively detect threats and protect our most critical digital assets.